

Template Demo Book: every trace on these pages was actually run

Each built-in template below is shown three ways: its real graph (drawn from the same coordinates the canvas uses), a full sim-mode run (the app's offline mode - real orchestration, canned agent text, zero model calls), and a live-mode run against the stalled-stub model from the integration tests, with the Axiom guard OFF and then ON.

Reproduce every number: `cd nodexloop && node tools/build-guard.mjs && node tools/demo-trace.mjs`, then `python docs/gen_nodexloop_template_demos_pdf.py`. The RNG is seeded, the network is disabled, and no model or API key is involved anywhere. Method: sim mode + stalled-stub live runs (tools/demo-trace.mjs), seeded RNG, no network, no model.

How to read the guard column

OFF -> ON compares model calls for the same stalled loop. A drop from the loop governor (LOOP_STALLED) is genuine savings. Where the drop comes from a PREFLIGHT BLOCK, the guard refused one of the template's own prompts before any call - for the prompt-injection sandbox that is the intended behaviour; for 'decision' and 'roi' it is a calibration finding we are reporting, not hiding: an adversarial persona ('devil's advocate', 'risk judge') trips a HARM pattern. Equal counts mean the guard found nothing to stop and cost zero extra calls.

All 23 at a glance

TEMPLATE	GROUP	SHAPE	SIM RUN	LIVE CALLS OFF -> ON
blog	build	6 nodes, 7 edges, 1 loop edge(s)	7 steps, 3 iter	7 -> 7 governor saved 0
code	build	5 nodes, 5 edges, 1 loop edge(s)	6 steps, 2 iter	7 -> 7 no intervention
trip	build	7 nodes, 8 edges	7 steps	4 -> 4 no intervention
synth	build	6 nodes, 7 edges, 1 loop edge(s)	8 steps, 3 iter	8 -> 8 governor saved 0
landing	build	5 nodes, 6 edges, 1 loop edge(s)	7 steps, 3 iter	7 -> 7 governor saved 0
data	build	5 nodes, 5 edges, 1 loop edge(s)	5 steps	2 -> 2 no intervention
debug	build	6 nodes, 6 edges, 1 loop edge(s)	7 steps, 2 iter	3 -> 3 no intervention
rl	build	8 nodes, 9 edges, 2 loop edge(s)	11 steps, 4 iter	4 -> 4 no intervention
robot	build	7 nodes, 8 edges, 1 loop edge(s)	8 steps, 2 iter	4 -> 4 no intervention
skill	build	7 nodes, 8 edges, 2 loop edge(s)	9 steps, 4 iter	3 -> 3 no intervention
bottleneck	company	7 nodes, 8 edges, 1 loop edge(s)	9 steps, 3 iter	10 -> 8 governor saved 2
drift	company	6 nodes, 7 edges, 1 loop edge(s)	8 steps, 3 iter	11 -> 7 governor saved 4
journey	company	6 nodes, 7 edges, 1 loop edge(s)	8 steps, 3 iter	10 -> 2 preflight block (calibration)
habit	life	6 nodes, 7 edges, 1 loop edge(s)	8 steps, 3 iter	10 -> 8 governor saved 2
decision	life	6 nodes, 7 edges, 1 loop edge(s)	8 steps, 2 iter	9 -> 0 preflight block (calibration)
timeaudit	life	6 nodes, 7 edges, 1 loop edge(s)	8 steps, 3 iter	10 -> 8 governor saved 2
roi	enterprise	7 nodes, 9 edges, 1 loop edge(s)	9 steps, 3 iter	10 -> 1 preflight block (calibration)
support	enterprise	6 nodes, 6 edges, 1 loop edge(s)	8 steps, 3 iter	11 -> 7 governor saved 4
selfheal	enterprise	6 nodes, 6 edges, 1 loop edge(s)	8 steps, 3 iter	14 -> 8 governor saved 6
invoice	enterprise	6 nodes, 6 edges, 1 loop edge(s)	9 steps, 4 iter	11 -> 7 governor saved 4
factcheck	meta	5 nodes, 6 edges, 1 loop edge(s)	7 steps, 3 iter	13 -> 7 governor saved 6
injection	meta	6 nodes, 7 edges, 1 loop edge(s)	8 steps, 3 iter	13 -> 7 governor saved 6
oscillation	meta	5 nodes, 6 edges, 2 loop edge(s)	6 steps, 3 iter	24 -> 6 governor saved 18

Build

The original ten - content, code, data and research loops.

01 - BUILD - blog

Blog post factory

Research -> Write -> Critic -> Publish. Iterates until quality threshold is met.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Topic input Received input - "User prompt + context"
Research agent Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Writer agent Drafting content - 1,428 words - 4 sections - tone matched to brief
Critic agent Scored iteration 1/3 - Score: 0.72 - "Weak transitions between key points"
Critic agent Scored iteration 2/3 - Score: 0.80 - "Conclusion does not follow from the analysis"
Critic agent Scored iteration 3/3 - Score: 0.90 - "Approved"
Final output Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.9 - 3 agent(s) - run completed

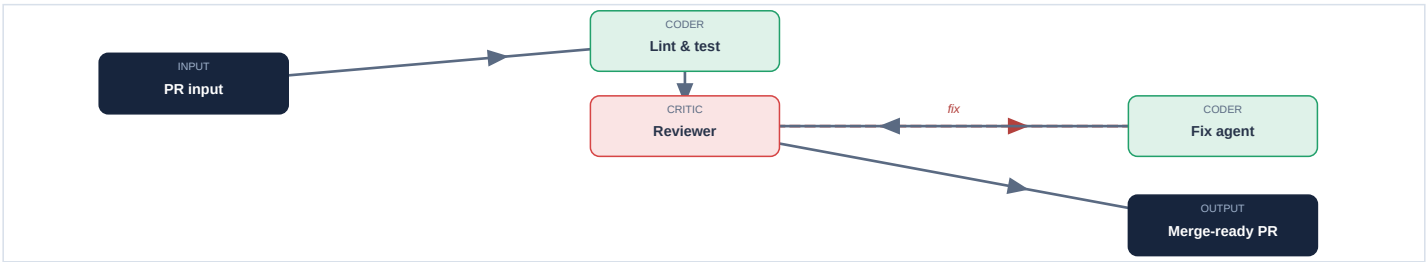
LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

7 model calls with the guard OFF -> 7 with it ON (nothing to stop; zero overhead)
[Critic agent] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 17 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

02 - BUILD - code

Code review cycle

Lint -> Test -> Review -> Fix. Automated PR review with iterative feedback.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

PR input Received input - "Diff + description"
Lint & test Writing implementation - 6 files changed - lint clean - 12 tests passing
Reviewer Scored iteration 1/3 - Score: 0.76 - "Claims need stronger sourcing"
Reviewer Scored iteration 2/3 - Score: 0.88 - "Approved"
Merge-ready PR Published result - Output document ready - see the Output tab
Fix agent Writing implementation - 4 files changed - lint clean - 19 tests passing

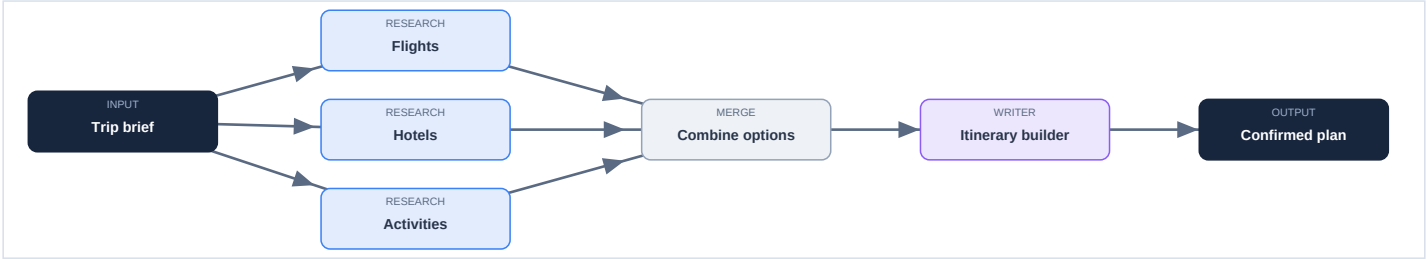
2 critic iteration(s) - final score 0.88 - 3 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

7 model calls with the guard OFF -> 7 with it ON (nothing to stop; zero overhead)
[axiom-guard] 17 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

Trip planner

Search flights -> Compare hotels -> Build itinerary -> Confirm bookings.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Trip brief	Received input - "User prompt + context"
Flights	Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Hotels	Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Activities	Searched 9 sources - 9 sources scanned - 6 relevant - key themes extracted to shared memory
Combine options	Combined streams - 3 input(s) merged into single context
Itinerary builder	Drafting content - 1,038 words - 6 sections - tone matched to brief
Confirmed plan	Published result - Output document ready - see the Output tab

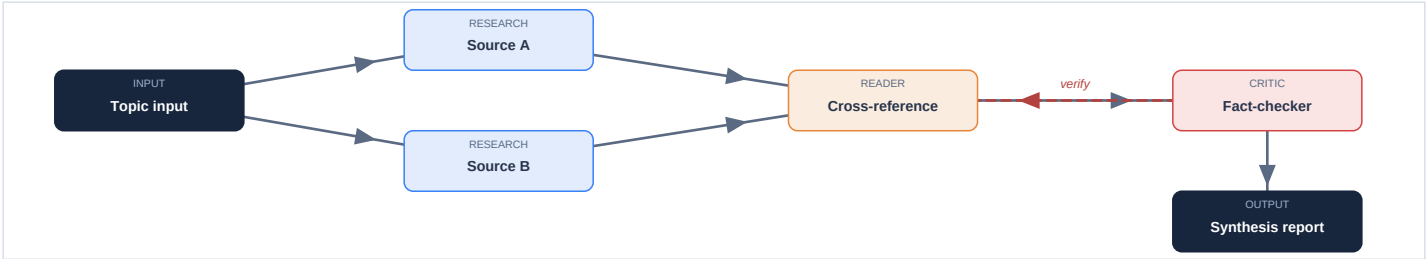
4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

4 model calls with the guard OFF -> 4 with it ON (nothing to stop; zero overhead)
[axiom-guard] 12 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

Research synthesizer

Multi-source research with cross-referencing and fact-checking loops.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Topic input	Received input - "User prompt + context"
Source A	Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Source B	Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Cross-reference	Reading and summarizing - 9 documents condensed - key points cross-referenced
Fact-checker	Scored iteration 1/3 - Score: 0.72 - "Claims need stronger sourcing"
Fact-checker	Scored iteration 2/3 - Score: 0.85 - "Good structure, needs tighter language"
Fact-checker	Scored iteration 3/3 - Score: 0.97 - "Approved"
Synthesis report	Published result - Output document ready - see the Output tab

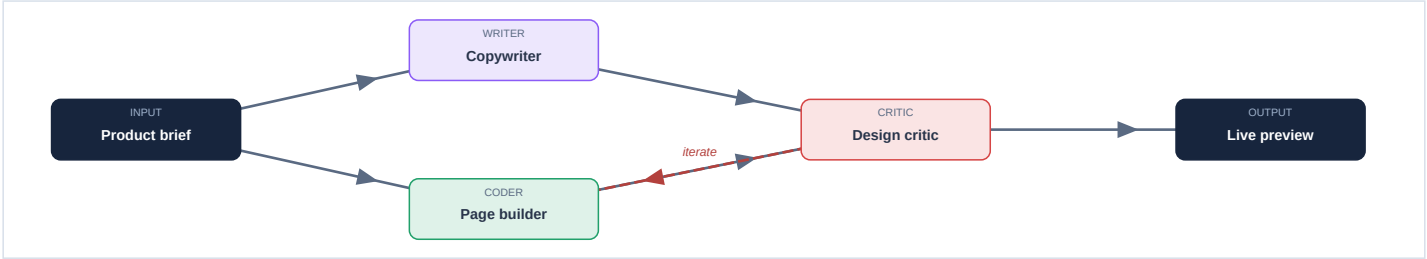
3 critic iteration(s) - final score 0.97 - 4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

8 model calls with the guard OFF -> 8 with it ON (nothing to stop; zero overhead)
[Fact-checker] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 20 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

Landing page builder

Copy -> Design -> Code -> Preview. Iterates on layout and messaging.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Product brief	Received input - "User prompt + context"
Copywriter	Drafting content - 1,170 words - 6 sections - tone matched to brief
Page builder	Writing implementation - 6 files changed - lint clean - 18 tests passing
Design critic	Scored iteration 1/3 - Score: 0.72 - "Weak transitions between key points"
Design critic	Scored iteration 2/3 - Score: 0.80 - "Conclusion does not follow from the analysis"
Design critic	Scored iteration 3/3 - Score: 0.90 - "Approved"
Live preview	Published result - Output document ready - see the Output tab

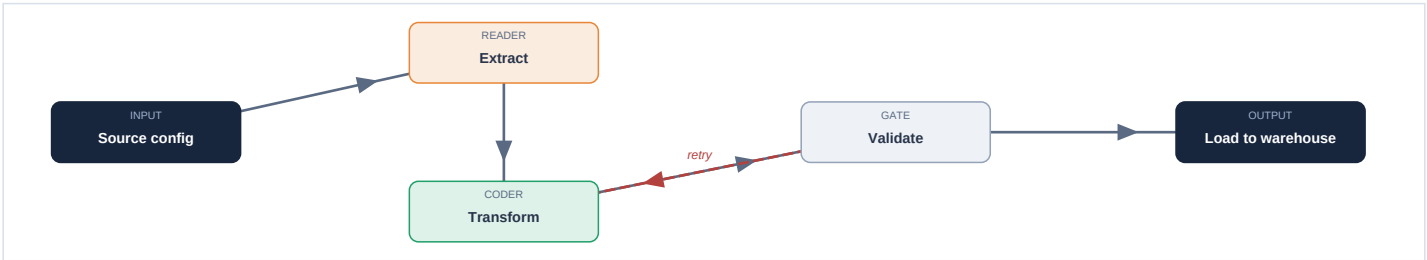
3 critic iteration(s) - final score 0.9 - 3 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

7 model calls with the guard OFF -> 7 with it ON (nothing to stop; zero overhead)
[Design critic] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 17 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

Data pipeline

Extract -> Transform -> Validate -> Load. With error recovery loops.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Source config	Received input - "User prompt + context"
Extract	Reading and summarizing - 11 documents condensed - key points cross-referenced
Transform	Writing implementation - 6 files changed - lint clean - 18 tests passing
Validate	Evaluated condition - Condition true -> continue
Load to warehouse	Published result - Output document ready - see the Output tab

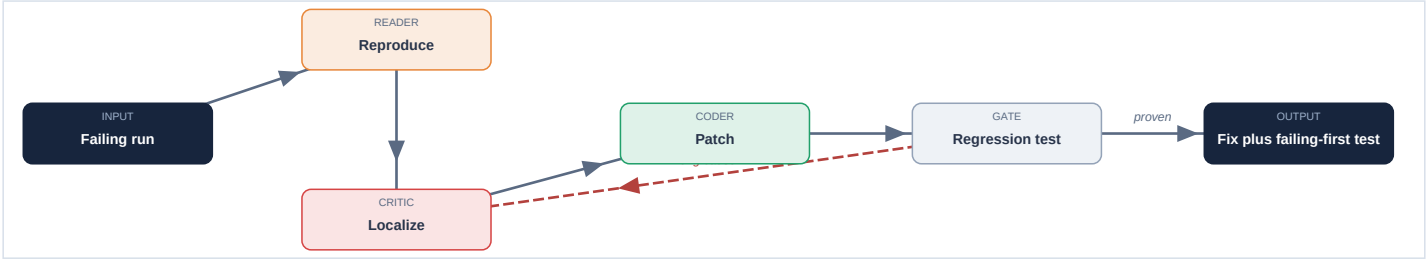
2 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

2 model calls with the guard OFF -> 2 with it ON (nothing to stop; zero overhead)
[axiom-guard] 6 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

Agent failure triage

Reproduce, localize, patch, prove. Turns "the agent did something weird" into a named cause and a test that fails before the fix and passes after it.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Failing run	Received input - "Run id, transcript, the step where it went wrong"
Reproduce	Reading and summarizing - 11 documents condensed - key points cross-referenced
Localize	Scored iteration 1/3 - Score: 0.76 - "Claims need stronger sourcing"
Localize	Scored iteration 2/3 - Score: 0.88 - "Approved"
Patch	Writing implementation - 4 files changed - lint clean - 19 tests passing
Regression test	Evaluated condition - Condition true -> continue
Fix plus failing-first tes	Published result - Output document ready - see the Output tab

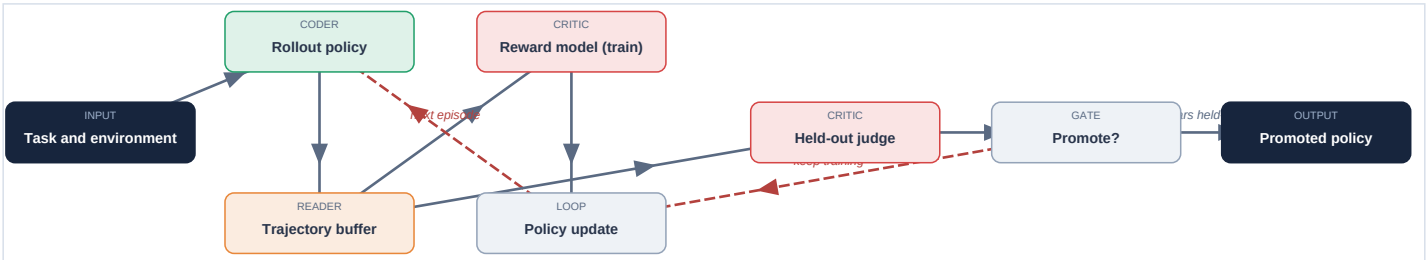
2 critic iteration(s) - final score 0.88 - 3 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

3 model calls with the guard OFF -> 3 with it ON (nothing to stop; zero overhead)
[axiom-guard] 9 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

Reward loop with a held-out judge

Rollout, reward, update, evaluate. The scorer that trains the policy is never the scorer that promotes it, so reward hacking shows up as a gap between the two.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Task and environment	Received input - "Objective, action space, episode budget"
Rollout policy	Writing implementation - 6 files changed - lint clean - 12 tests passing
Trajectory buffer	Reading and summarizing - 11 documents condensed - key points cross-referenced
Reward model (train)	Scored iteration 1/8 - Score: 0.72 - "Weak transitions between key points"
... 4 more steps ...	
Held-out judge	Scored iteration 2/3 - Score: 0.86 - "Approved"
Promote?	Evaluated condition - Condition true -> continue
Promoted policy	Published result - Output document ready - see the Output tab

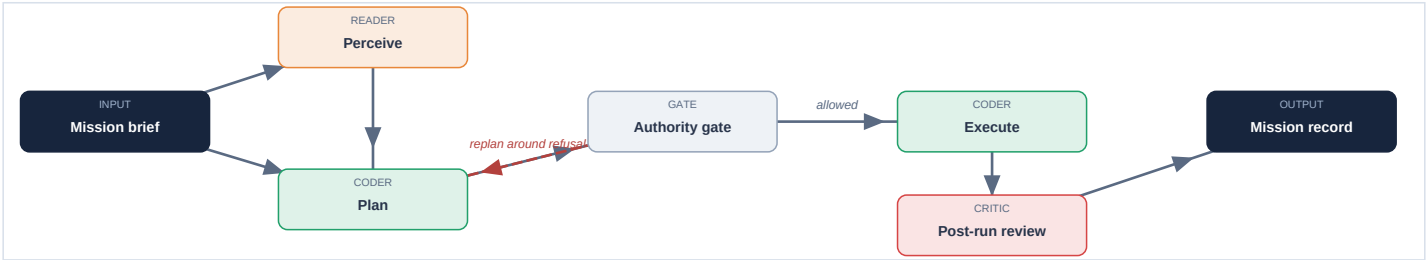
4 critic iteration(s) - final score 0.86 - 4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

4 model calls with the guard OFF -> 4 with it ON (nothing to stop; zero overhead)
[axiom-guard] 12 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

Robot task authority

Perceive, plan, authority gate, act, review. The plan is checked against the trust level of the robot before a motor turns, and every refusal is recorded. Pairs with the Fleet sandbox.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Mission brief	Received input - "Goal, robot id, trust level, allowed zones"
Perceive	Reading and summarizing - 11 documents condensed - key points cross-referenced
Plan	Writing implementation - 6 files changed - lint clean - 18 tests passing
Authority gate	Evaluated condition - Condition true -> continue
Execute	Writing implementation - 4 files changed - lint clean - 27 tests passing
Post-run review	Scored iteration 1/3 - Score: 0.74 - "Examples are generic - add specifics"
Post-run review	Scored iteration 2/3 - Score: 0.85 - "Approved"
Mission record	Published result - Output document ready - see the Output tab

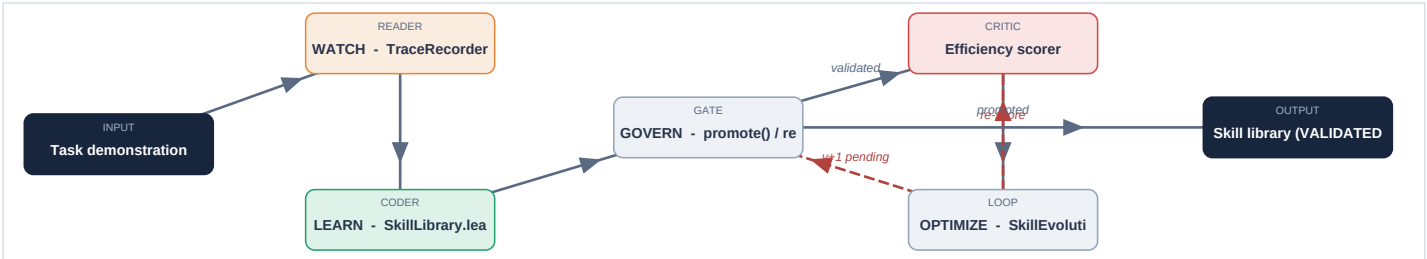
2 critic iteration(s) - final score 0.85 - 4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

4 model calls with the guard OFF -> 4 with it ON (nothing to stop; zero overhead)
[axiom-guard] 13 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

Skill acquisition pipeline

WATCH -> LEARN -> OPTIMIZE -> GOVERN. An agent learns a reusable skill from a signed task demonstration - but nothing becomes usable until a named human promotes it. Experimental.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Task demonstration	Received input - "One real task execution - task + domain"
WATCH - TraceRecorder	Reading and summarizing - 11 documents condensed - key points cross-referenced
LEARN - SkillLibrary.lea	Writing implementation - 6 files changed - lint clean - 18 tests passing
GOVERN - promote() / rej	Evaluated condition - Condition true -> continue
... 2 more steps ...	
Efficiency scorer	Scored iteration 3/5 - Score: 0.94 - "Weak transitions between key points"
Efficiency scorer	Scored iteration 4/5 - Score: 0.99 - "Approved"
Skill library (VALIDATED o	Published result - Output document ready - see the Output tab

4 critic iteration(s) - final score 0.99 - 3 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

3 model calls with the guard OFF -> 3 with it ON (nothing to stop; zero overhead)
[axiom-guard] 9 check(s), 0 blocked, 0 loop iteration(s) stopped early (~0 estimated tokens saved)

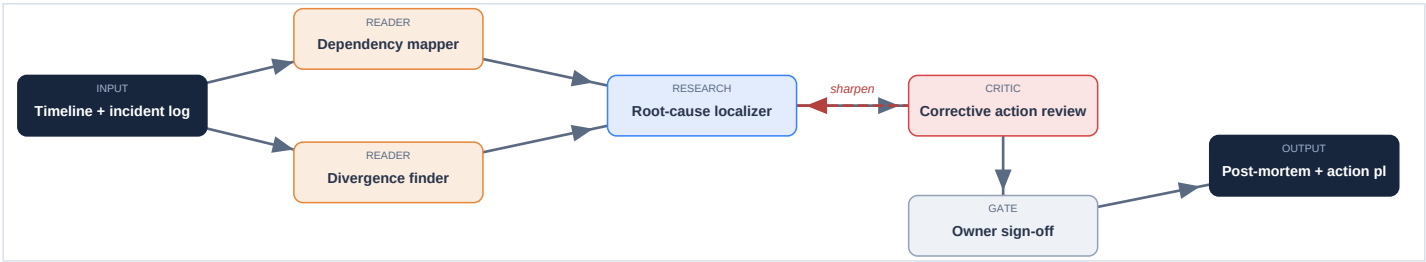
Debug your company

Org processes as loop graphs: delivery, strategy, customer journeys.

11 - COMPANY - bottleneck

Delivery bottleneck localizer

Pin the plan against what actually happened and step to the first point they diverge. Treats a stalled delivery like a hung process: find the wait, not the symptom.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Timeline + incident log	Received input - "Paste the project timeline: planned milestones, actual delivery dates, who handed off to whom, and any blockers raised."
Dependency mapper	Reading and summarizing - 11 documents condensed - key points cross-referenced
Divergence finder	Reading and summarizing - 11 documents condensed - key points cross-referenced
Root-cause localizer	Searched 9 sources - 9 sources scanned - 6 relevant - key themes extracted to shared memory
... 2 more steps ...	
Corrective action reviewer	Scored iteration 3/4 - Score: 0.97 - "Approved"
Owner sign-off	Evaluated condition - Condition true -> continue
Post-mortem + action plan	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.97 - 4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

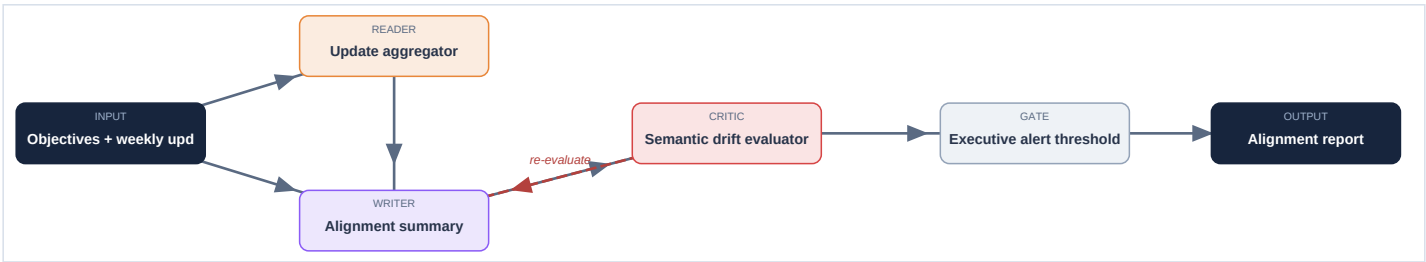
10 model calls with the guard OFF -> 8 with it ON

[Corrective action reviewer] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 21 check(s), 0 blocked, 1 loop iteration(s) stopped early (~34 estimated tokens saved)

12 - COMPANY - drift

Strategy drift detector

Scores what teams actually did against what the quarter said they would do. A falling score is drift - the governor sees it before the quarterly review does.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Objectives + weekly update	Received input - "Paste the quarterly objectives, then the last few weeks of team updates."
Update aggregator	Reading and summarizing - 11 documents condensed - key points cross-referenced
Alignment summary	Drafting content - 1,428 words - 4 sections - tone matched to brief
Semantic drift evaluator	Scored iteration 1/5 - Score: 0.72 - "Weak transitions between key points"
Semantic drift evaluator	Scored iteration 2/5 - Score: 0.80 - "Conclusion does not follow from the analysis"
Semantic drift evaluator	Scored iteration 3/5 - Score: 0.90 - "Approved"
Executive alert threshold	Evaluated condition - Condition true -> continue
Alignment report	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.9 - 3 agent(s) - run completed

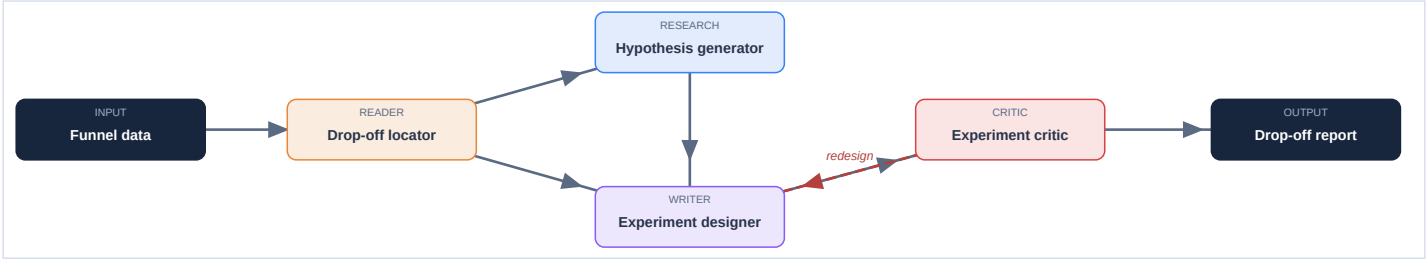
LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

11 model calls with the guard OFF -> 7 with it ON

[Semantic drift evaluator] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 18 check(s), 0 blocked, 2 loop iteration(s) stopped early (~68 estimated tokens saved)

Customer drop-off debugger

Walk the funnel node by node to the step that loses people, generate competing explanations, and design an experiment that could actually settle it.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Funnel data	Received input - "Paste the funnel: each step, how many entered it, how many continued, and how long they spent there."
Drop-off locator	Reading and summarizing - 11 documents condensed - key points cross-referenced
Hypothesis generator	Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Experiment designer	Drafting content - 1,140 words - 5 sections - tone matched to brief
Experiment critic	Scored iteration 1/4 - Score: 0.72 - "Claims need stronger sourcing"
Experiment critic	Scored iteration 2/4 - Score: 0.85 - "Good structure, needs tighter language"
Experiment critic	Scored iteration 3/4 - Score: 0.97 - "Approved"
Drop-off report	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.97 - 4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

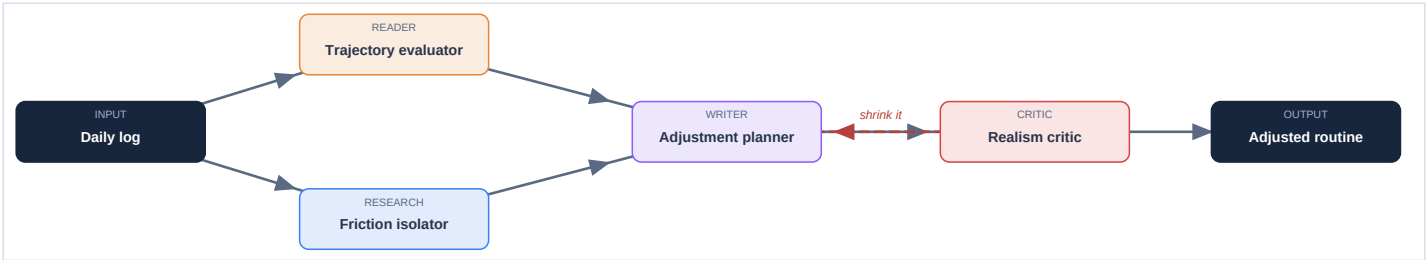
10 model calls with the guard OFF -> 2 with it ON
[axiom-guard] Preflight blocked Experiment critic - HARM (harm:1)
[axiom-guard] 1 node(s) blocked before any model call - their branches will be skipped
Calibration finding: the template's own adversarial persona trips a HARM pattern - the ON count reflects a skipped branch, not savings.

Debug your life

Personal loops: habits, decisions, where the hours actually go.

Habit trajectory tracker

Why the routine collapsed in week three. Effort plotted against progress - a flat trajectory under sustained effort is the finding, and the fix lowers friction rather than demanding more discipline.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Daily log	Received input - "Paste your log: the habit you are trying to keep, which days you managed it, at what time, and what happened on the days"
Trajectory evaluator	Reading and summarizing - 11 documents condensed - key points cross-referenced
Friction isolator	Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Adjustment planner	Drafting content - 1,140 words - 5 sections - tone matched to brief
Realism critic	Scored iteration 1/4 - Score: 0.72 - "Claims need stronger sourcing"
Realism critic	Scored iteration 2/4 - Score: 0.85 - "Good structure, needs tighter language"
Realism critic	Scored iteration 3/4 - Score: 0.97 - "Approved"
Adjusted routine	Published result - Output document ready - see the Output tab

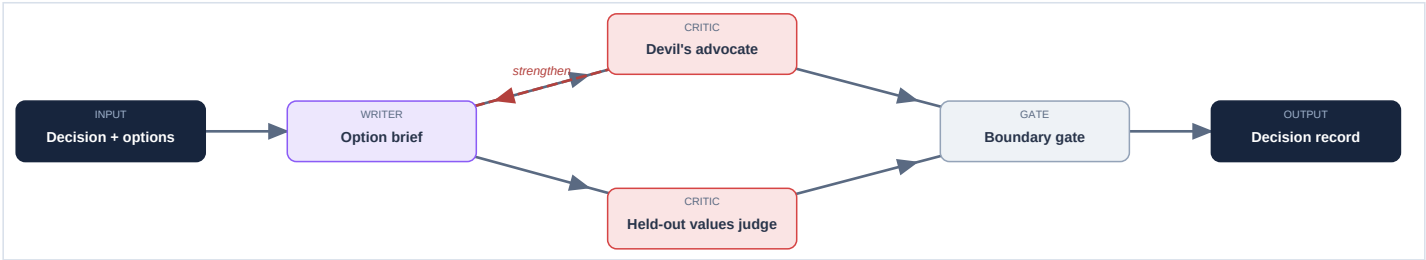
3 critic iteration(s) - final score 0.97 - 4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

10 model calls with the guard OFF -> 8 with it ON
[Realism critic] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 21 check(s), 0 blocked, 1 loop iteration(s) stopped early (~34 estimated tokens saved)

Decision preflight

Run a big decision past a devil's advocate and a held-out values judge that sits outside the loop, so it cannot be argued with. A crossed boundary stops at the gate.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Decision + options	Received input - "Describe the decision, the options you are weighing, the constraints that are real, and the values or boundaries you do "
Option brief	Drafting content - 1,170 words - 6 sections - tone matched to brief
Devil's advocate	Scored iteration 1/4 - Score: 0.76 - "Claims need stronger sourcing"
Devil's advocate	Scored iteration 2/4 - Score: 0.88 - "Approved"
Held-out values judge	Scored iteration 1/2 - Score: 0.69 - "Claims need stronger sourcing"
Held-out values judge	Scored iteration 2/2 - Score: 0.80 - "Weak transitions between key points"
Boundary gate	Evaluated condition - Condition false -> fallback path
Decision record	Published result - Output document ready - see the Output tab

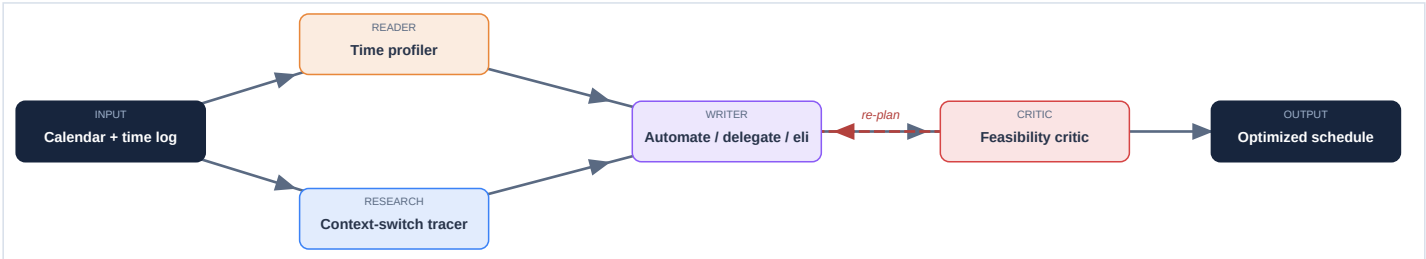
2 critic iteration(s) - final score 0.8 - 3 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

9 model calls with the guard OFF -> 0 with it ON
[axiom-guard] Preflight blocked Devil's advocate - HARM (harm:1)
[axiom-guard] 1 node(s) blocked before any model call - their branches will be skipped
Calibration finding: the template's own adversarial persona trips a HARM pattern - the ON count reflects a skipped branch, not savings.

Attention leak debugger

Profile where the hours go and trace what each interruption costs downstream. Every high-drain item is automated, delegated or eliminated - keeping it is not an option.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Calendar + time log	Received input - "Paste a representative two weeks: meetings, recurring commitments, and roughly where the rest of the time went."
Time profiler	Reading and summarizing - 11 documents condensed - key points cross-referenced
Context-switch tracer	Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Automate / delegate / elim	Drafting content - 1,140 words - 5 sections - tone matched to brief
Feasibility critic	Scored iteration 1/4 - Score: 0.72 - "Claims need stronger sourcing"
Feasibility critic	Scored iteration 2/4 - Score: 0.85 - "Good structure, needs tighter language"
Feasibility critic	Scored iteration 3/4 - Score: 0.97 - "Approved"
Optimized schedule	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.97 - 4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

10 model calls with the guard OFF -> 8 with it ON
[Feasibility critic] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 21 check(s), 0 blocked, 1 loop iteration(s) stopped early (~34 estimated tokens saved)

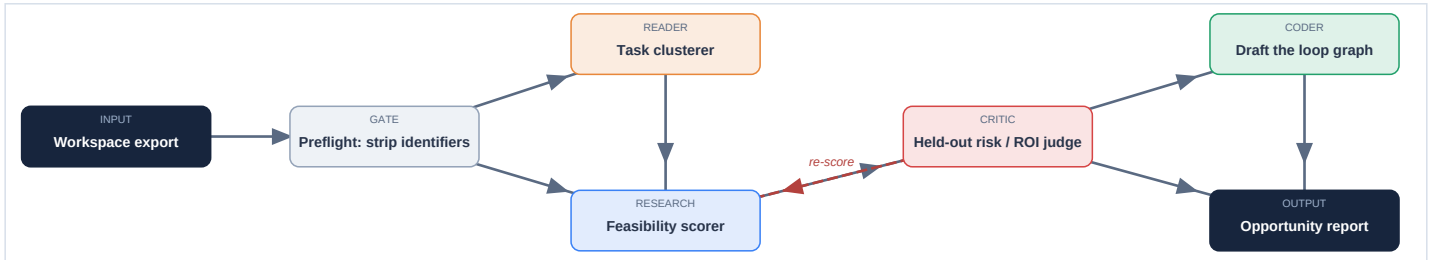
Enterprise operations

Support, self-healing CI, finance and AI-ROI auditing.

17 - ENTERPRISE - roi

AI opportunity and ROI auditor

Point it at your SOPs, tickets or wiki. It clusters the work that repeats, scores hours saved against tokens spent, has a held-out judge attack the case, and drafts a .loop.yaml for the best candidate.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Workspace export	Received input - "Paste or attach internal material: standard operating procedures, a ticket export, wiki pages, or call summaries."
Preflight: strip identifiers	Evaluated condition - Condition true -> continue
Task clusterer	Reading and summarizing - 5 documents condensed - key points cross-referenced
Feasibility scorer	Searched 7 sources - 7 sources scanned - 4 relevant - key themes extracted to shared memory
... 2 more steps ...	
Held-out risk / ROI judge	Scored iteration 3/4 - Score: 0.94 - "Approved"
Draft the loop graph	Writing implementation - 5 files changed - lint clean - 38 tests passing
Opportunity report	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.94 - 4 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

10 model calls with the guard OFF -> 1 with it ON

[axiom-guard] Preflight blocked Held-out risk / ROI judge - HARM (harm:1)

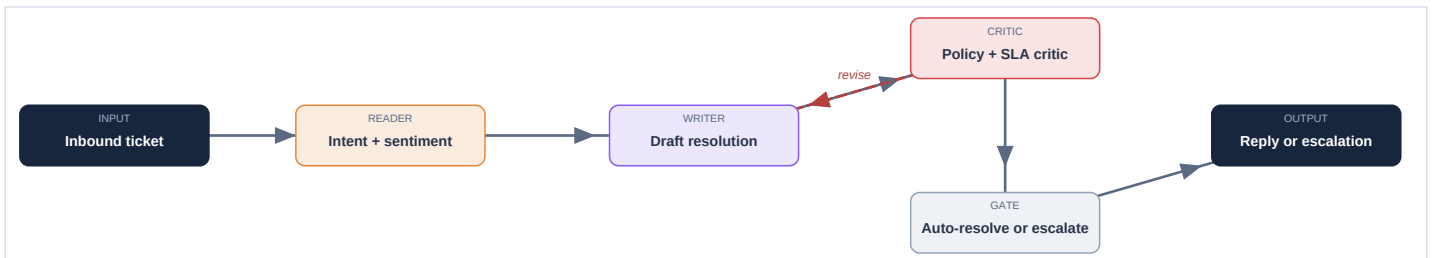
[axiom-guard] 1 node(s) blocked before any model call - their branches will be skipped

Calibration finding: the template's own adversarial persona trips a HARM pattern - the ON count reflects a skipped branch, not savings.

18 - ENTERPRISE - support

Support escalation loop

Classify, draft, check against policy and SLA, resolve or escalate. Catches the refund nobody authorized, and the agent rewriting the same email without adding a fix.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Inbound ticket	Received input - "Paste the ticket: what the customer said, their plan or tier, and any relevant history."
Intent + sentiment	Reading and summarizing - 11 documents condensed - key points cross-referenced
Draft resolution	Drafting content - 1,428 words - 4 sections - tone matched to brief
Policy + SLA critic	Scored iteration 1/5 - Score: 0.72 - "Weak transitions between key points"
Policy + SLA critic	Scored iteration 2/5 - Score: 0.80 - "Conclusion does not follow from the analysis"
Policy + SLA critic	Scored iteration 3/5 - Score: 0.90 - "Approved"
Auto-resolve or escalate	Evaluated condition - Condition true -> continue
Reply or escalation	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.9 - 3 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

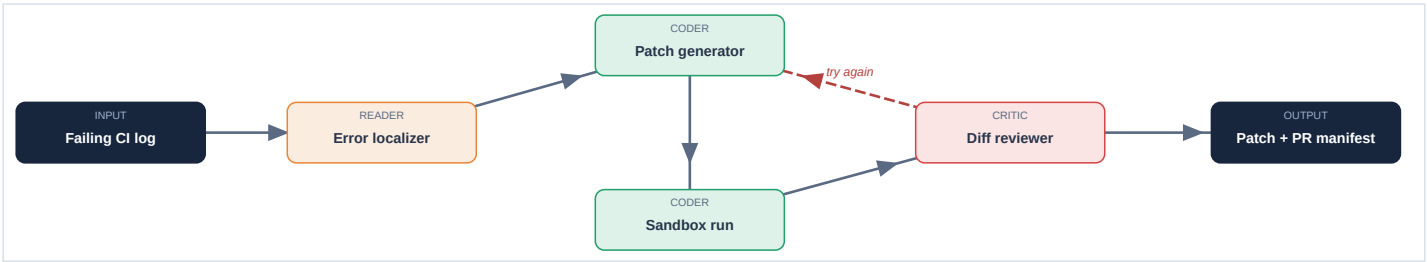
11 model calls with the guard OFF -> 7 with it ON

[Policy + SLA critic] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)

[axiom-guard] 18 check(s), 0 blocked, 2 loop iteration(s) stopped early (-68 estimated tokens saved)

Self-healing CI loop

Failing build to reviewed patch: localize the first real error, fix the cause, prove it in a sandbox, and block anything that only silences the symptom.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Failing CI log	Received input - "Paste the failing CI output, the test that fails, and the change that preceded it."
Error localizer	Reading and summarizing - 11 documents condensed - key points cross-referenced
Patch generator	Writing implementation - 6 files changed - lint clean - 18 tests passing
Sandbox run	Writing implementation - 2 files changed - lint clean - 28 tests passing
Diff reviewer	Scored iteration 1/6 - Score: 0.72 - "Claims need stronger sourcing"
Diff reviewer	Scored iteration 2/6 - Score: 0.85 - "Good structure, needs tighter language"
Diff reviewer	Scored iteration 3/6 - Score: 0.97 - "Approved"
Patch + PR manifest	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.97 - 4 agent(s) - run completed

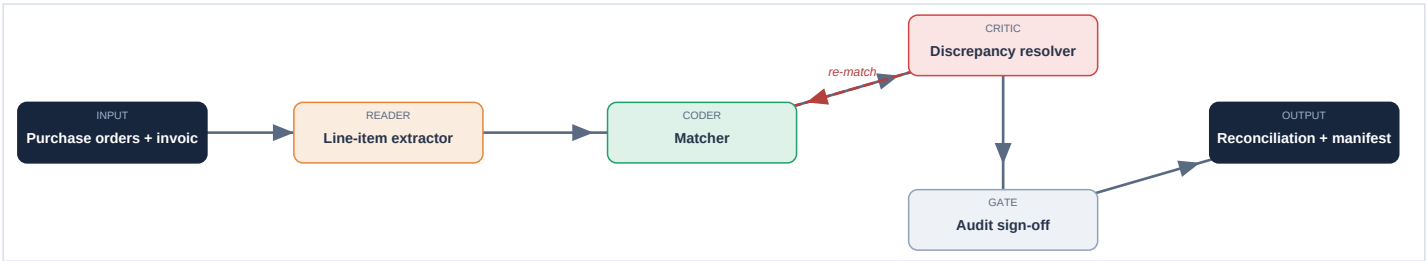
LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

14 model calls with the guard OFF -> 8 with it ON

[Diff reviewer] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 21 check(s), 0 blocked, 3 loop iteration(s) stopped early (~102 estimated tokens saved)

Invoice reconciliation and audit

Extract line items, match both sides, explain every difference or flag it. Nothing is inferred to close a gap, and unexplained differences stop at sign-off.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Purchase orders + invoices	Received input - "Paste or attach the purchase orders and the invoices to reconcile."
Line-item extractor	Reading and summarizing - 11 documents condensed - key points cross-referenced
Matcher	Writing implementation - 6 files changed - lint clean - 18 tests passing
Discrepancy resolver	Scored iteration 1/5 - Score: 0.72 - "Weak transitions between key points"
... 2 more steps ...	
Discrepancy resolver	Scored iteration 4/5 - Score: 0.98 - "Approved"
Audit sign-off	Evaluated condition - Condition true -> continue
Reconciliation + manifest	Published result - Output document ready - see the Output tab

4 critic iteration(s) - final score 0.98 - 3 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

11 model calls with the guard OFF -> 7 with it ON

[Discrepancy resolver] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 18 check(s), 0 blocked, 2 loop iteration(s) stopped early (~68 estimated tokens saved)

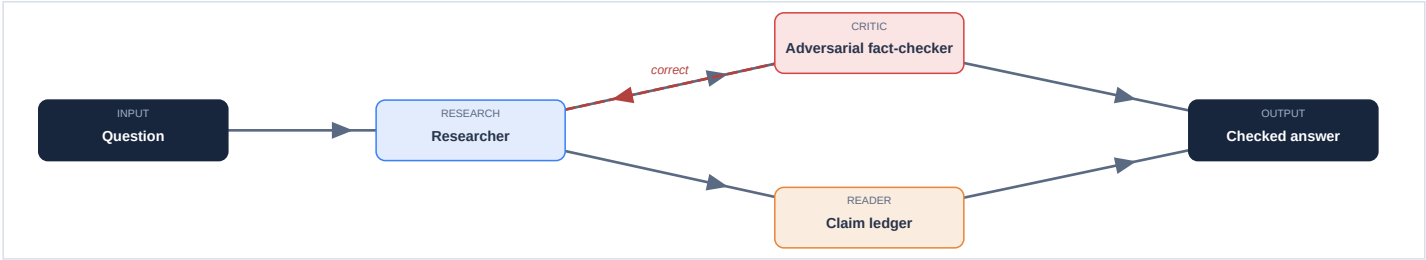
Meta

Loops that test the loops: hallucination, prompt injection, drift.

21 - META - factcheck

Hallucination bench

A researcher paired with a fact-checker that assumes every claim is false until the answer supports it. Break on the step where the citations start being invented.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Question	Received input - "State the question the researcher should answer, and any sources it is allowed to rely on."
Researcher	Searched 11 sources - 11 sources scanned - 8 relevant - key themes extracted to shared memory
Adversarial fact-checker	Scored iteration 1/6 - Score: 0.76 - "Claims need stronger sourcing"
Adversarial fact-checker	Scored iteration 2/6 - Score: 0.88 - "Weak transitions between key points"
Adversarial fact-checker	Scored iteration 3/6 - Score: 0.97 - "Approved"
Claim ledger	Reading and summarizing - 9 documents condensed - key points cross-referenced
Checked answer	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.97 - 3 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

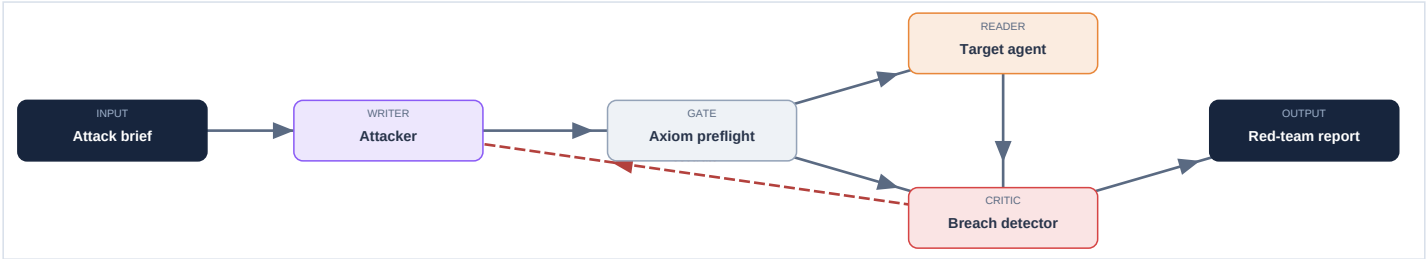
13 model calls with the guard OFF -> 7 with it ON

[Adversarial fact-checker] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 18 check(s), 0 blocked, 3 loop iteration(s) stopped early (~102 estimated tokens saved)

22 - META - injection

Prompt injection sandbox

An attacker loop hammering an agent with overrides, encoded payloads and indirect injection - with the local Axiom preflight in front, so a blocked attempt costs no model call at all.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Attack brief	Received input - "Describe the agent under test, the instructions it is meant to hold to, and what would count as a breach."
Attacker	Drafting content - 1,170 words - 6 sections - tone matched to brief
Axiom preflight	Evaluated condition - Condition true -> continue
Target agent	Reading and summarizing - 7 documents condensed - key points cross-referenced
Breach detector	Scored iteration 1/6 - Score: 0.71 - "Weak transitions between key points"
Breach detector	Scored iteration 2/6 - Score: 0.82 - "Conclusion does not follow from the analysis"
Breach detector	Scored iteration 3/6 - Score: 0.94 - "Approved"
Red-team report	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.94 - 3 agent(s) - run completed

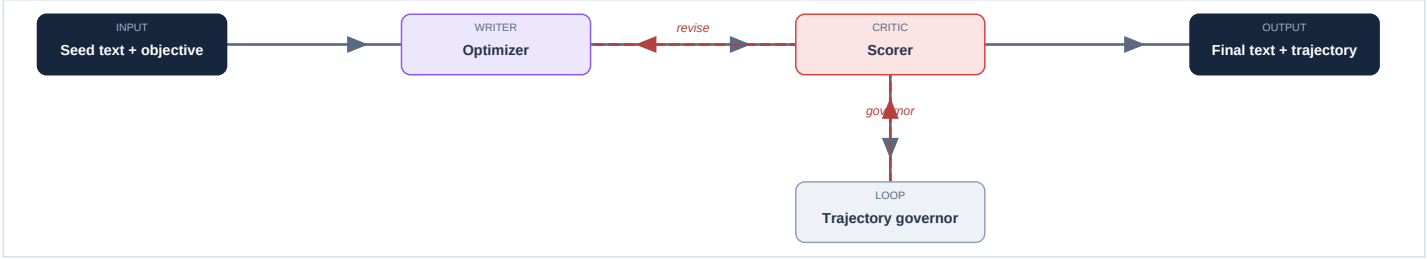
LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

13 model calls with the guard OFF -> 7 with it ON

[Breach detector] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[axiom-guard] 18 check(s), 0 blocked, 3 loop iteration(s) stopped early (~102 estimated tokens saved)

Drift and oscillation detector

An optimizer and a scorer, run long enough to misbehave. Watch the trajectory flatten or turn non-monotonic, and watch the governor stop it before the budget goes.



SIM RUN - REAL ORCHESTRATION, ZERO MODEL CALLS

Seed text + objective	Received input - "Paste the text to optimize and state precisely what would make it better."
Optimizer	Drafting content - 1,170 words - 6 sections - tone matched to brief
Scorer	Scored iteration 1/12 - Score: 0.76 - "Claims need stronger sourcing"
Scorer	Scored iteration 2/12 - Score: 0.88 - "Weak transitions between key points"
Scorer	Scored iteration 3/12 - Score: 0.97 - "Approved"
Final text + trajectory	Published result - Output document ready - see the Output tab

3 critic iteration(s) - final score 0.97 - 2 agent(s) - run completed

LIVE RUN, STALLED-STUB MODEL - GUARD OFF vs ON

24 model calls with the guard OFF -> 6 with it ON

[Scorer] Loop stopped by Axiom guard - LOOP_STALLED: draft unchanged for 2 iterations (novelty 0)
[Trajectory governor] loop node passed context through